Portugal
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# Guilherme Lopes

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## Work Experience

**Protocol Engineer** **May 2025 - February 2026**
*Somnia Foundation / Virtual Society Foundation – Role concluded as core client milestones were delivered and team scope narrowed*

- **Contributed** broadly to a high-performance C++ networking daemon by resolving core data encoding bugs, implementing RPC API endpoints, and engineering compiler optimizations for the virtual machine that achieved a 15% execution speed-up.
- **Optimized** database internals for heavy workloads—specifically improving data compression during massive data writes—yielding a 10% performance increase under heavy load.
- **Architected** an external data ingestion pipeline, building robust C++ off-chain worker nodes to fetch and serialize external API and LLM data into the system state.

**Systems Engineer** **June 2024 - May 2025**
*Community Labs / Astro Labs – Role concluded as company funding runway ended*

- **Architected** and built a cross-network stablecoin transfer system connecting independent distributed networks to securely process multi-network asset transfers.
- **Implemented** core system components including external data pipelines to facilitate decentralized data synchronization.
- **Secured** distributed message validation by deploying MPC-based signing nodes (C++) on Trusted Execution Environment (TEE) hardware to prevent replay attacks and ensure key integrity.

**Software Engineering Intern** **June 2023 - September 2023**
*Nethermind*

- **Contributed** to a distributed Go client library by developing and fixing automated tests, ensuring system reliability and code quality.

**Distributed Systems Researcher** **September 2022 - September 2023**

- **Researched** and implemented distributed consensus algorithms for a master's thesis (research grant awarded).
- **Developed** a modular consensus library and a live-testing tool in OCaml to easily integrate and evaluate the performance metrics (e.g., throughput, latency) of custom protocols on a distributed node.

**Software Engineer (Freelance / Contract)** **October 2019 - September 2022**

- **Managed** an independent freelancing business, collaborating directly with clients to define technical requirements and deliver end-to-end software solutions.
- **Developed** and deployed full-stack web applications, microservices, and internal automation tools (e.g., ticket systems, inventory management) using Rust, Go, and Python.

## Education

**Master of Science (MSc) in Computer Science** **2021 - 2023**
*Blockchain, Consensus Protocols, Compilers*
*University of Beira Interior (Top 3 in class)*

**Bachelor of Science in Computer Science** **2018 - 2021**
*Distributed Systems (Best in Class)*
*University of Beira Interior*

## Relevant Projects

**Arrow Distributed Directory Protocol and Distributed Database / Filesystem**
- Implemented a distributed directory protocol node in Go with a live visualization tool for network state and access control, and later created a distributed database and filesystem using Rust and Libp2p to enhance efficient data storage and retrieval in a peer-to-peer network.

**Web Applications & Backend Services (Rust, Go, TypeScript)**
- Built various full-stack applications, backend services, and automated tooling utilizing Rust, TypeScript, and Go.

**9p Network Filesystem Protocol (Elixir)**
- Implemented a robust 9p filesystem server protocol in Elixir, allowing remote directories to be mounted and served concurrently over TCP.

**On-disk Key Value Database**
- Engineered a key-value disk database in C, utilizing B-trees for optimized data management and access.

**Language Design / Compiler**
- Coded a Go-based Lisp interpreter, and an OCaml-based Pascal to Assembly (X_86) compiler.

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## Skills

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| **Programming** | Rust, C/C++, Go, Python, Solidity, TypeScript/JavaScript |
| **Technical Skills** | Linux, Git, Web Development, Libp2p, CI/CD, Containerization, Databases (SQL/NoSQL), REST APIs, Data Structures & Algorithms |
| **Languages** | Portuguese (native speaker), English (fluent) |

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## Interests

**Technical Interests**
Distributed Systems, Consensus Algorithms, Blockchains, Cryptography, Functional Programming, Compilers, Formal Verification, Operating Systems, Computer Security

**Hobbies & Past Activities**
Music (Saxophone, 5 yrs), Basketball (3 yrs), Fishing, Beekeeping, Farming, Cycling

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## Awards and Events

**Solana Breakpoint Conferences 2022 and 2023** **November 2023**
*Received Student Fellowship Ticket @ Lisbon, Portugal/Amesterdam, Netherlands*

**ICPC SWERC 2023** **February 2023**
*South Eastern Europe Programming Competition @ Politecnico Milano, Milan, Italy*

**MIUP 2022** **October 2022**
*Portuguese National Programming Competition @ Coimbra University, Coimbra, Portugal*